HW #2

ALGEBRAIC NUMBER THEORY, GU4043; INSTRUCTOR: GYUJIN OH

Question 1. Let $f(X) = X^3 - 15X - 20$.

(1) Show that $f(X)$ is irreducible in $\mathbb{Q}[X]$.

(2) Let $K = \mathbb{Q}(\alpha)$, where α is a root of $f(X)$. Use the techniques of Exercise 1.7 to show that

$$[\mathcal{O}_K : \mathbb{Z}[\alpha]] = 1 \text{ or } 2.$$

Hint. To exclude 3, you need to consider $f(X + a)$ instead for an appropriate a, just like Exercise 1.7(4).

(3) Show that there is no subring $\mathbb{Z}[\alpha] \subset R \subset K$ such that $[R : \mathbb{Z}[\alpha]] = 2$. Deduce that $\mathcal{O}_K = \mathbb{Z}[\alpha]$. [1]

Hint. Such an R must be a $\mathbb{Z}[\alpha]$-submodule of $\frac{1}{2}\mathbb{Z}[\alpha]$. When can it be a **ring**?

Proof. (1) Follows immediately from the Eisenstein's irreducibility criterion (Exercise 1.4(1)) with $p = 5$.

(2) Note that $[\mathcal{O}_K : \mathbb{Z}[\alpha]]^2$ divides $D(1, \alpha, \alpha^2)$, which is computed for cubic polynomials in Exercise 1.2:

$$D(1, \alpha, \alpha^2) = -27\cdot(-20)^2 - 4\cdot(-15)^3 = -27\cdot 400 + 60\cdot 225 = -10800 + 13500 = 2700 = 2^2\cdot 3^3\cdot 5^2$$

Therefore, $[\mathcal{O}_K : \mathbb{Z}[\alpha]]$ divides $2 \cdot 3 \cdot 5 = 30$.

As $f(X)$ is Eisenstein at 5, by Exercise 1.7(3), we know that $[\mathcal{O}_K : \mathbb{Z}[\alpha]]$ is coprime to 5. We also note that $f(X) \equiv X^3 + 1 \pmod 3$, so there is a chance that $f(X-1)$ is Eisenstein at 3; indeed,

$$f(X-1) = (X-1)^3 - 15(X-1) - 20 = (X^3 - 3X^2 + 3X - 1) + (-15X + 15) - 20 = X^3 - 3X^2 - 12X - 6,$$

which is Eisenstein at 3. Since $\alpha + 1$ is a root of $f(X - 1)$ and $K = \mathbb{Q}(\alpha) = \mathbb{Q}(\alpha + 1)$, $[\mathcal{O}_K : \mathbb{Z}[\alpha + 1]] = [\mathcal{O}_K : \mathbb{Z}[\alpha]]$ is coprime to 3. Thus, we know that $[\mathcal{O}_K : \mathbb{Z}[\alpha]]$ is either 1 or 2.

(3) If $r \in R$, then $2r \in \mathbb{Z}[\alpha]$, as $R/\mathbb{Z}[\alpha]$ is additively an order 2 group. Therefore, $R \subset \frac{1}{2}\mathbb{Z}[\alpha]$. Thus, the order two group $R/\mathbb{Z}[\alpha]$ is represented by $\{1, \beta = \frac{a+b\alpha+c\alpha^2}{2}\}$ for $a, b, c \in \{0, 1\}$, $(a, b, c) \neq (0, 0, 0)$.

We can exclude many triples (a, b, c) by using the fact that $\beta^2, \beta^3, \cdots$ are elements of $R \subset \frac{1}{2}\mathbb{Z}[\alpha] = \left\{\frac{x+y\alpha+z\alpha^2}{2} \mid x, y, z \in \mathbb{Z}\right\}$. Note that

$$\beta^2 = \frac{a^2 + 2ba\alpha + (b^2 + 2ac)\alpha^2 + 2bc\alpha^3 + c^2\alpha^4}{4}$$

[1] This can be done more systematically using Dedekind's index criterion, which we will learn later.

Using $\alpha^3 = 15\alpha + 20$, we have

$$\begin{aligned}\beta^2 &= \frac{a^2 + 2ba\alpha + (b^2 + 2ac)\alpha^2 + 2bc(15\alpha + 20) + c^2(15\alpha^2 + 20\alpha)}{4} \\ &= \frac{(a^2 + 40bc) + (2ba + 30bc + 20c^2)\alpha + (b^2 + 2ac + 15c^2)\alpha^2}{4}.\end{aligned}$$

Thus, $a^2 + 40bc$, $2ba + 30bc + 20c^2$, $b^2 + 2ac + 15c^2$ are all even numbers. The second number $2ba + 30bc + 20c^2$ is obviously even, and the other two being even means a^2 is even and $b^2 + 15c^2$ is even. Therefore, $a = 0$ and $b \equiv c \pmod 2$. As $(a, b, c) \neq (0, 0, 0)$, This implies that only $(a, b, c) = (0, 1, 1)$ is the only remaining possibility.

For $\beta = \frac{\alpha+\alpha^2}{2}$,

$$\beta^2 = \frac{40 + 50\alpha + 16\alpha^2}{4} = \frac{20 + 25\alpha + 8\alpha^2}{2}$$

which is indeed in $\frac{1}{2}\mathbb{Z}[\alpha]$. Let's try β^3:

$$\begin{aligned}\beta^3 &= \left(\frac{20 + 25\alpha + 8\alpha^2}{2}\right)\left(\frac{\alpha + \alpha^2}{2}\right) \\ &= \frac{20\alpha + 45\alpha^2 + 33\alpha^3 + 8\alpha^4}{4} \\ &= \frac{20\alpha + 45\alpha^2 + 33(15\alpha + 20) + 8(15\alpha^2 + 20\alpha)}{4} \\ &= \frac{660 + (20 + 495 + 160)\alpha + (45 + 120)\alpha^2}{4} \\ &= \frac{660 + 675\alpha + 165\alpha^2}{4}\end{aligned}$$

As this is not in $\frac{1}{2}\mathbb{Z}[\alpha]$, we have a contradiction, and the assumption that there is a ring $\mathbb{Z}[\alpha] \subset R \subset K$ with $[R : \mathbb{Z}[\alpha]] = 2$ is false. Thus, $[\mathcal{O}_K : \mathbb{Z}[\alpha]]$ is never 2, and we conclude that $\mathcal{O}_K = \mathbb{Z}[\alpha]$.

□

Question 2. Let K, L be number fields over $\mathbb{Q}$ (**not necessarily Galois**) such that $K \cap L = \mathbb{Q}$.

(1) If $(\operatorname{disc}(K), \operatorname{disc}(L)) = 1$, show that $\mathcal{O}_{KL} = \mathcal{O}_K\mathcal{O}_L$ and $\operatorname{disc}(KL) = \operatorname{disc}(K)^{[L:\mathbb{Q}]} \operatorname{disc}(L)^{[K:\mathbb{Q}]}$.

Hint. Modify the proof of Proposition 1.35; use Proposition 1.16(6).

(2) In general, show that

$$\operatorname{lcm}(\operatorname{disc}(K), \operatorname{disc}(L)) \mid \operatorname{disc}(KL) \mid \operatorname{disc}(K)^{[L:\mathbb{Q}]} \operatorname{disc}(L)^{[K:\mathbb{Q}]}.$$

Proof. (1) The first part of the proof of Proposition 1.35, showing that $\mathcal{O}_K\mathcal{O}_L = \mathcal{O}_{KL}$, can be verbatim used by setting $\tau_1, \cdots, \tau_n$ to be the K-embeddings $KL \hookrightarrow \overline{\mathbb{Q}}$; then the proof shows that $\operatorname{disc}(L)\beta_j$ is an algebraic integer in K, so $\operatorname{disc}(L)\beta_j \in \mathcal{O}_K$.

The second part can be verbatim used if one shows the following group theory lemma (choose a Galois closure of $KL/\mathbb{Q}$ and work inside it, using Proposition 1.16(6)).

Lemma 1. *Let G be a finite group, and let $H_1, H_2 \leq G$ be two subgroups (not necessarily normal). Let $H = H_1 \cap H_2$. Suppose that $H_1H_2 = G$. Then, you can find coset representatives $G/H_1 = \{g_1, \cdots, g_m\}$ and $G/H_2 = \{h_1, \cdots, h_n\}$ such that G/H is represented by $\{g_1h_1, \cdots, g_1h_n, \cdots, g_mh_1, \cdots, g_mh_n\}$.*

Proof. First note that $[G : H] = [G : H_1][G : H_2]$, as $\frac{[G:H]}{[G:H_1]} = [H_1 : H] = [H_1H_2 : H_2] = [G : H_2]$; here H_1H_2 is a union of right cosets of H_2, and $[H_1 : H] = [H_1H_2 : H_2]$ as H_1 acts transitively on H_1H_2/H_2 with the stabilizer of H_2 being $H = H_1 \cap H_2$.

Choose any set of coset reprsentatives $G/H_1 = \{g_1, \cdots, g_m\}$. Also, choose any set of coset representatives $H_1/H = \{h_1, \cdots, h_n\}$, where $n = [H_1 : H] = [G : H_2]$. These can be used as coset representatives of G/H_2 by the above paragraph. To show that these coset representatives work, it suffices to show that, if $g_ih_jH = g_{i'}h_{j'}H$, then $(i, j) = (i', j')$ (we just showed that the order of the finite set G/H is mn). As h_k's are all elements of H_1, it follows that $g_ih_jH = g_{i'}h_{j'}H$ implies $g_iH_1 = g_{i'}H_1$, which implies that $i = i'$. Thus, $h_jH = h_{j'}H$, which then implies that $j = j'$ as desired. □

(2) Immediate from Theorem 1.33 and $\mathcal{O}_K\mathcal{O}_L \subset \mathcal{O}_{KL}$; the computation done in (1) shows that, in general, $D(\mathcal{O}_K\mathcal{O}_L) = \operatorname{disc}(K)^{[L:\mathbb{Q}]} \operatorname{disc}(L)^{[K:\mathbb{Q}]}$.

□

Question 3. Let p be an odd prime. In this Question, we will identify the unique quadratic subfield of $\mathbb{Q}(\zeta_p)$ using another method (related to the **Gauss sums** that will come later in the course).

(1) Let K be the unique quadratic subfield of $\mathbb{Q}(\zeta_p)$. Show that

$$K = \mathbb{Q}(\alpha), \quad \alpha := \sum_{1 \leq a \leq p-1,\ a \text{ is a quadratic residue mod } p} \zeta_p^a$$

(2) Show that $\operatorname{Tr}_{K/\mathbb{Q}}(\alpha) = -1$.

Hint. Let $\sigma : K \to K$ be the unique nontrivial element of $\operatorname{Gal}(K/\mathbb{Q})$. What is $\sigma(\alpha)$? What is $\alpha + \sigma(\alpha)$?

(3) Recall the **Legendre symbol** from Introduction,

$$\left(\frac{a}{p}\right) := \begin{cases} 1 & \text{if } a \text{ is a quadratic residue mod } p, \\ -1 & \text{if } a \text{ is not a quadratic residue mod } p. \end{cases}$$

Show that, for $a, b \not\equiv 0 \pmod{p}$, $\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right)$.

(4) Show that

$$(\alpha - \sigma(\alpha))^2 = \begin{cases} p & \text{if } -1 \text{ is a quadratic residue mod } p, \\ -p & \text{if } -1 \text{ is not a quadratic residue mod } p. \end{cases}$$

Hint. Write $(\alpha - \sigma(\alpha))^2$ as a double sum using the Legendre symbols, and massage it.

(5) Deduce that

$$K = \begin{cases} \mathbb{Q}(\sqrt{p}) & \text{if } -1 \text{ is a quadratic residue mod } p, \\ \mathbb{Q}(-\sqrt{p}) & \text{if } -1 \text{ is not a quadratic residue mod } p. \end{cases}$$

(6) Deduce that -1 is a quadratic residue mod p if and only if $p \equiv 1 \pmod{4}$.

Proof. (1) Let $H \subset \mathrm{Gal}(\mathbb{Q}(\zeta_p)/\mathbb{Q})$ be the index 2 subgroup. If we identify $\mathrm{Gal}(\mathbb{Q}(\zeta_p)/\mathbb{Q}) \cong (\mathbb{Z}/p\mathbb{Z})^\times$, $\sigma_a \leftrightarrow a$, where $\sigma_a(\zeta_p) = \zeta_p^a$, then H is precisely the subgroup of quadratic residues mod p. Therefore, the elements of $\mathbb{Q}(\zeta_p)$ fixed by H is precisely the $\mathbb{Q}$-linear combination of 1 and α. This shows that $K = \mathbb{Q}(\alpha)$.

(2) Note that $\mathrm{Gal}(K/\mathbb{Q}) \cong \mathrm{Gal}(\mathbb{Q}(\zeta_p)/\mathbb{Q})/H$, and this group, an order 2 group, has a non-trivial element represented by σ_x for any x which is not a quadratic residue. Therefore,

$$\sigma(\alpha) = \sum_{1\le a\le p-1,\ a \text{ is a quadratic residue mod } p} \zeta_p^{ax} = \sum_{1\le a\le p-1,\ a \text{ is } \underline{\text{not}} \text{ a quadratic residue mod } p} \zeta_p^{a}.$$

Therefore,

$$\alpha + \sigma(\alpha) = \sum_{1\le a\le p-1} \zeta_p^a = -1$$

(3) Note that $(\mathbb{Z}/p\mathbb{Z})^\times$ is a cyclic group of order $(p-1)$. This means that there is a (multiplicative) generator $x \in (\mathbb{Z}/p\mathbb{Z})^\times$; namely, $\{1, x, \cdots, x^{p-2}\}$ subsumes all $p-1$ elements of $(\mathbb{Z}/p\mathbb{Z})^\times$. From this, it is easy to see that an element $y \in (\mathbb{Z}/p\mathbb{Z})^\times$ is a quadratic residue if and only if $y = x^k$ for even k. Namely,

$$\left(\frac{x^k}{p}\right) = (-1)^k.$$

From this description, the multiplicativity is obvious.

(4) Note that $\alpha - \sigma(\alpha)$ can be expressed as

$$\alpha - \sigma(\alpha) = \sum_{1\le a\le p-1} \left(\frac{a}{p}\right) \zeta_p^a.$$

Therefore,

$$(\alpha - \sigma(\alpha))^2 = \sum_{1\le a,b\le p-1} \left(\frac{a}{p}\right) \zeta_p^a \left(\frac{b}{p}\right) \zeta_p^b = \sum_{1\le a,b\le p-1} \left(\frac{ab}{p}\right) \zeta_p^{a+b}.$$

We can reindex the sum using $b = ax$:

$$(\alpha - \sigma(\alpha))^2 = \sum_{1\le a,x\le p-1} \left(\frac{a^2x}{p}\right) \zeta_p^{a+ax} = \sum_{1\le x\le p-1} \left(\frac{x}{p}\right) \sum_{1\le a\le p-1} \zeta_p^{a(x+1)}$$

The inner sum is -1 if $x+1 \not\equiv 0 \pmod p$, and $p-1$ if $x+1 \equiv 0 \pmod p$. Thus

$$(\alpha - \sigma(\alpha))^2 = \sum_{1\le x\le p-2} \left(\frac{x}{p}\right) \cdot (-1) + \left(\frac{p-1}{p}\right) \cdot (p-1).$$

Note that the number of quadratic residues is exactly the half of $(\mathbb{Z}/p\mathbb{Z})^\times$, so $\sum_{1\le x\le p-1} \left(\frac{x}{p}\right) = 0$. Therefore

$$(\alpha - \sigma(\alpha))^2 = \sum_{1\le x\le p-2} \left(\frac{x}{p}\right) \cdot (-1) + \left(\frac{p-1}{p}\right) \cdot (p-1) = p\left(\frac{-1}{p}\right).$$

This is exactly the description we have been looking for.

(5) From (2) and (4), we see that $\sigma = \frac{-1\pm\sqrt{(\alpha-\sigma(\alpha))^2}}{2}$, from which the description follows.

(6) This follows from comparing (5) and Exercise 1.6.

$\square$